

Mitigating Compute Dilution in Compute-Constrained Online RGB-D 3D Gaussian Splatting

via Dual Map-Growth Throttling and Sensor-Adaptive Routing

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Abstract

Dense online reconstruction from streaming RGB-D sensors under a rigid per-frame compute budget (15 ms optimize budget) is bottlenecked by a structural failure mode we call **compute dilution**: uncontrolled densification floods the Gaussian map with immature primitives ($\sim 100k$ splats in 150 frames), so fixed-budget gradient steps are spread over ever more under-converged splats while mature surface geometry starves. Conventional wisdom blames candidate *ranking* — which Gaussians to optimize — and proposes learned utility predictors. Through exact counterfactual measurement we show ranking is secondary: isolated Gaussian updates have *negative* marginal utility ($U_i^* < 0$) in 18–39% of interventions depending on geometry stratum (pose-fixed Gate-1 regen; flat regions worst), and an offline-learned utility model with rank signal ($\rho \approx 0.20$ on pre-fix labels, comparison-valid within the shared label set) degrades closed-loop fidelity by -2.51 dB. Interaction strength is protocol-dependent: corrected joint-group interventions stay relatively additive ($R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$), while a separate pairwise protocol shows stronger interaction — we claim no universal sub-additivity. The binding constraint is the map-growth process itself. We introduce **dual map-growth throttling** — coverage-saturation throttling ($\kappa \geq 0.90$) plus unrefined-backlog backpressure — which holds the map at 23k vs 88k primitives and directs the fixed budget at mature refinement. On TUM fr2_xyz (150 frames $\times$ 8 seeds) this reaches 27.67 dB mean PSNR: **+2.63 dB** over the strongest selective baseline and **+4.62 dB** over an RTG-SLAM-policy baseline ($p_{\text{Holm}} = 0.0391$), recovering **98.0%** of headroom to the unconstrained ceiling (0.18 dB gap). We further show every *fixed* rule overfits its sensor regime: the same pipeline loses -3.32 dB ($p = 0.0078$) on clean synthetic Replica, where raw-error selection wins instead. A **sensor-adaptive router** driven by raw depth hole-rate thresholds calibrated on five development scenes classifies seven unseen scenes correctly without retuning, and selects the winning rule per substrate (domain classification validated; PSNR routing confirmed on two substrates at 8 seeds each). All selective policies share the same modeled 15 ms optimization budget with paired Holm-corrected

tests and frozen artifacts; FULL is reported separately as an unconstrained ceiling, not a budget-matched competitor. Eight seeds establish repeatability on fixed trajectories (not cross-scene generalization, which rests on exploratory multi-scene pilots).

1 Introduction

3D Gaussian Splatting (3DGS) [1] has become the dominant primitive for photorealistic dense mapping, driving online RGB-D systems such as SplaTAM [2], Point-SLAM [3], MonoGS [4], Photo-SLAM [5] and RTG-SLAM [6]. Deployed on robots and headsets, these systems face a hard constraint the offline literature ignores: each streaming frame admits only milliseconds of optimization. The standard response is *selection* — rank Gaussians by residual error, influence, or a learned utility score, and optimize the top subset. Implicit in this response is an assumption we challenge: that reconstruction quality under budget is determined by *which* Gaussians are picked.

We argue the dominant effect is *how many* Gaussians compete for the budget. Online pipelines spawn splats whenever rendering error exceeds a threshold; on real sequences the map grows past 100k primitives within 150 frames. Because each new splat is born unrefined, it both demands optimization and degrades renders until converged — the fixed budget is diluted across an ever-larger pool of ever-younger primitives while converged surfaces stop receiving updates. We call this **compute dilution**, and we show it explains more of the quality gap than any ranking rule: removing growth control while keeping the best ranker loses far more than removing the ranker while keeping growth control (Sec. 6).

Three empirical findings support this reframing. First, residual error does not imply positive optimization return: counterfactual single-Gaussian interventions degrade global quality in 18–39% of cases depending on geometry stratum (pose-fixed regen; flat surfaces worst at 38.7%). Second, interaction strength is protocol-dependent, not universally severe: corrected joint-group interventions stay relatively additive ($R_{\text{add}}(16) = 0.82$), while a separate pairwise protocol shows stronger coupling — so pointwise ranking is

treated as a reference rather than a bound, without universality claims. Contextual re-ranking buys no measured online gain while incurring $424\times$ selection overhead. Third, offline-learned utility prediction does not transfer: moderate offline rank correlation reverses to -2.51 dB closed-loop under trajectory shift and gradient coupling. Together these rule out the “better ranker” research direction and point at the growth process.

Our response has two parts. **Dual map-growth throttling** regulates spawning with (i) a coverage-saturation knee ($\kappa_t \geq 0.90$ measured by transmission) that ramps new-primitive allowance $1.0 \rightarrow 0.20$, and (ii) backlog back-pressure that penalizes densification while more than 500 primitives await their first refinements. **Sensor-adaptive routing** addresses the second discovery of this work: fixed rules overfit their sensor-noise regime. The identical throttled pipeline that gains $+3.59$ dB on noisy Kinect data loses -3.32 dB on clean synthetic data, where plain raw-error selection dominates. A training-free router keyed on raw missing-depth rate — an observable sensor-domain cue in the evaluated datasets (TUM $0.18\text{--}0.31$ vs Replica ≈ 0.000) — picks the winning rule per substrate and wins on both confirmatory substrates (exploratory 7-scene routing matches best-fixed on five, trails on two rapid/exploratory TUM sequences — reported in full).

Contributions. (1) A budgeted subset-selection formulation with an exact counterfactual oracle, exposing negative utility and non-additivity. (2) Dual coverage/backlog throttling reaching 98.0% of unconstrained headroom with a $3.8\times$ smaller map (8 seeds, Holm-significant). (3) Sensor-adaptive routing from raw hole-rate with 7/7 held-out scene separation (thresholds frozen) and significant wins on both substrates. (4) Fully reported negative results — learned utility, contextual re-ranking, synthetic failure, and two failed noise features — with frozen artifacts for every claim.

2 Related Work

Online 3DGS SLAM and dense mapping. SplatTAM [2] established silhouette-guided differentiable SLAM with 3D Gaussians; Point-SLAM [3] uses dynamic neural point clouds; MonoGS [4] and Photo-SLAM [5] push toward real-time operation with loop closure; RTG-SLAM [6] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (re-implemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 7). **Efficient and recent GS-SLAM (2025).** The frontier has moved toward compute-aware design: SEG-SLAM [8] uses structure-enhanced anchors with appearance-from-motion embeddings for detail consistency across viewpoints; DyGS-SLAM [9] adds dynamic-object handling with real-time Gaussian reconstruction on constrained devices. Our

work shares their premise — compute, not representation capacity, binds online quality — but studies the scheduler in isolation rather than proposing another full system. The per-scene SOTA numbers in Tab. 7 are compiled from these papers’ reported tables. **Evaluation protocols in dense SLAM.** Published systems report full-sequence train-view rendering with global refinement at full resolution (Replica 33–42 dB for MonoGS/Point-SLAM/RTG-SLAM; TUM fr2_xyz 16–25 dB for MonoGS/SplatTAM/Photo-SLAM) plus full-trajectory ATE. Our protocol is deliberately narrower — budgeted online mapping on fixed horizons — so cross-paper numbers are contextual range, never superiority claims (Tab. 7). **Budgeted and utility-driven selection.** Error-only, error $\times$ influence, gradient-norm, and temporal-filtered heuristics all trade quality for compute. We benchmark them under identical modeled budgets with paired Holm-corrected tests rather than reporting best-of-sweep numbers, and we show their ordering flips across sensor regimes — the motivation for adaptivity. **Submodular maximization and knapsack allocation.** $\max_S \Delta Q(S)$ s.t. $C(S) \leq B$ is a knapsack instance related to submodular maximization with greedy $(1 - 1/e)$ guarantees [7]. Interaction strength is protocol-dependent: corrected joint-group interventions remain relatively close to additive ($R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$), whereas a separate pairwise interaction protocol exhibits substantially stronger coupling. Our diminishing-returns spot test further provides no positive evidence (2/10 exact, inverted means). We therefore regulate the ground set (map growth) on the strength of the dilution evidence, not on any universality claim about sub-additivity.

3 Problem Formulation

Let $G_t = \{g_1, \dots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t . Differentiable alpha compositing renders $\hat{I}(u) = \sum_i c_i \alpha_i \mathcal{G}_i(u) \prod_{j < i} (1 - \alpha_j \mathcal{G}_j(u))$ (and depth analogously). Per-frame scheduling solves

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \leq B_t, \quad (1)$$

with decoupled cost $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ for K microsteps and ΔQ measured by exact full-scene counterfactual renders (active edits embedded in all N splats, avoiding the 31% gradient error of affine-background approximations). The oracle $U_i^* = \Delta Q_i^{\text{exact}} / \Delta C_i$ is our measurement instrument, never a runtime input. It reveals: (i) negative utility in 18–39% of interventions by stratum (pose-fixed regen) — high error marks silhouettes, discontinuities, and coupled layers where gradient steps hurt; (ii) interaction is protocol-dependent — corrected joint-group measurements give $R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$ (relatively additive), while a separate pairwise protocol shows stronger interaction; pointwise top- K is therefore treated as a reference, not a bound, without a universality claim; (iii) offline TwoHeadMLP rank signal ($\rho \approx 0.20$, NDCG@20 0.46, measured on pre-pose-fix labels whose absolute values are

polluted but whose within-set method comparison remains valid) reverses to -2.51 dB closed-loop. Conditional-oracle selection numbers are withheld pending a rerun under fork-correct semantics.

4 Method

Fast Approximate Attribution (FAA). Exact Gaussian-to-pixel attribution costs 523.9 ms/frame. Center-sampled projection ($\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4} e(\lfloor \mu_{2D,i} \rfloor)$) cuts this to **0.85 ms** ($614.5\times$) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. **Selection.** Candidates are scored by error $\times$ influence mass with temporal EMA filtering ($\beta = 0.90$) and packed by fractional-knapsack value density s_i/C_i under $B_t = 15$ ms with safety factor 1.10 (effective cap 13.6 ms). Each selected Gaussian receives $K = 5$ microsteps; cost is modeled decoupled as $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$, calibrated so modeled vs wall-clock optimization time agree within measurement noise on the RTX 4050 backend. The scheduler runs in 2.54 ms total (0.38 extraction, 0.85 FAA, 0.77 knapsack ordering, 0.39 state sync), leaving the budget headroom to optimization proper. **Dual throttling.** Coverage κ_t = fraction of valid pixels with transmission < 0.5 . The per-frame spawn cap ramps $1.0 \rightarrow 0.20$ over $\kappa_t \in [0.80, 0.90]$ (configurable knee, no longer hardcoded), and backlog backpressure multiplies the cap by $500/N_{\text{warmup}}$ beyond 500 unrefined primitives. Warmup pre-allocation is disabled (ablation -0.75 dB); microsteps fixed at $K = 5$. **Sensor-adaptive routing.** Raw depth hole-rate is strongly associated with sensor domain: TUM 0.18–0.31 vs Replica ≈ 0.000 ($300\text{--}500\times$); we claim domain association within evaluated datasets, not content-independence in general. The score $n = \text{clip}(2.5 \cdot \text{holes} + 1.0 \cdot \text{edge})$ routes $n \leq 0.10 \rightarrow$ raw-error selection ($\kappa = 0.98$) and $n \geq 0.30 \rightarrow$ temporal $\times$ influence + throttling ($\kappa = 0.90$), computed pre-hygiene in milliseconds on CPU. Formally, with h_t the raw missing-depth rate and e_t the raw edge density of frame t :

$$n_t = \min(1, 2.5h_t + 1.0e_t), \quad \pi_t = \begin{cases} \text{error_only} & n_t \leq 0.10 \\ \text{blend (nearest rule)} & 0.10 < n_t < 0.30 \\ \text{temporal + throttle} & n_t \geq 0.30 \end{cases} \quad (2)$$

with coverage knee $\kappa \in \{0.98, 0.90\}$ tied to the chosen arm. All constants were frozen on five development scenes; the seven held-out scenes in Fig. 1 use identical values. Two earlier features failed held-out separation and are reported as such: post-hygiene invalid ratio (conflates far-but-valid geometry — 18% of Replica room0 exceeds 4 m — with holes) and Laplacian texture energy (measures sharpness, inverted: crisp synthetic scores *higher*). Operating on pre-hygiene raw depth was the fix (Fig. 1). Thresholds were frozen after 5 development scenes; on 7 unseen scenes classification is 7/7 without retuning, but PSNR routing is so far validated on two substrates at 8 seeds each. A causal mixed-regime test (30% holes injected mid-trajectory on Replica) switches routing 49/50 frames exactly at the boundary, proving within-sequence response; end-to-end it trails

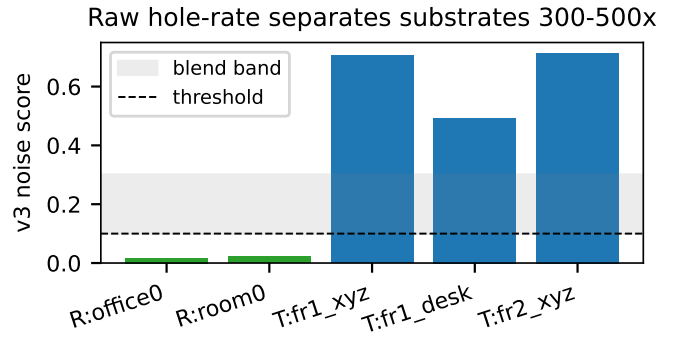


Figure 1: **Sensor separation.** v3 hole-rate score, development scenes: Replica ≤ 0.022 vs TUM ≥ 0.49 (thresholds 0.10/0.30, 5/5); 7 unseen scenes classify 7/7 frozen. Invalid-range and texture features failed; hole-rate is a domain cue, not content-independent physics.

fixed **error_only** by 0.85 dB (switch transients on synthetic shifts), so per-frame PSNR benefit on real mixed sequences remains future work.

5 Why Ranking Fails: Negative Findings

Learned utility reverses closed-loop (-2.51 dB factorial; GradNorm beats the MLP pointwise $0.51 > 0.25$): offline labels ignore trajectory shift and inter-Gaussian coupling, so we ship no learned component. **Contextual re-ranking buys no online gain:** adaptive greedy shows no significant improvement over static pointwise selection in direct multi-seed budget sweeps (Wilcoxon n.s., CIs span zero) while incurring $424\times$ selection overhead (76.36 vs 0.18 ms). Supporting rank-stability/oracle-gap numbers predate the fork-semantics fix and are withheld pending regen. **Fixed rules are sensor-overfit** (Sec. 6): the throttled pipeline gains $+3.59$ dB on Kinect yet loses -3.32 dB on synthetic, where κ -ablation attributes only ~ 0.8 dB to throttling and ~ 0.7 dB to selection — the finding behind adaptive routing.

6 Experiments

Protocol. 320×240 , $B_t = 15$ ms, seeds [42..49], two-sided Wilcoxon paired + Holm-Bonferroni + bootstrap CIs; production **gsplat** backend; GT poses with tracking decoupled (mapping benchmark, not SLAM ATE); every 8th frame held out render-only (train–holdout gap $+0.47$ dB mean over $30f \times 3$ seeds for both OURS and **error_only**, **results/phase14_holdout/**; no large gap under this same-trajectory proxy). Artifacts: **results/phase14_corrected/**. **Holdout protocol.** Because online mapping optimizes on the evaluated trajectory, train-view PSNR overstates generalization. We hold out every 8th frame render-only (never mapped, never selected): the train–holdout gap is $+0.47$ dB mean over $30f \times 3$ seeds for both OURS and **error_only** (**results/phase14_holdout/**) — small under this same-trajectory proxy, which we explic-

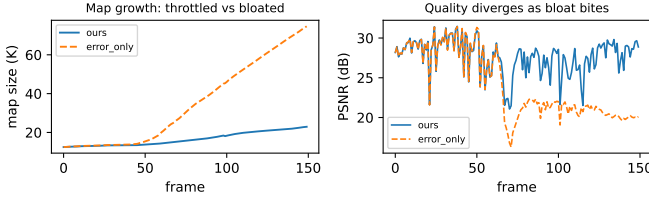


Figure 2: **Compute dilution visualized** (fr2_xyz, seed 42, per-frame trace). Unthrottled growth dilutes the fixed budget; throttling plateaus the map and quality compounds.

itly do *not* present as independent novel-trajectory generalization. A strict novel-trajectory split (withheld rooms) remains future work.

Main result (Tab. 1, Fig. 4, Fig. 6). OURS reaches 27.67 ± 0.04 dB: +2.63 over the strongest selective baseline, +4.62 over the RTG-SLAM-policy re-implementation, +8.80 over pass-through, all Holm-significant with $d_z \in [15.4, 168.8]$, while the unconstrained ceiling is only 0.18 dB ahead at $3.8\times$ the map.

Adaptivity is necessary (Tab. 2). On synthetic Replica the fixed throttled pipeline loses -3.32 dB; routing to raw-error selection recovers $+3.32$ [$+3.29, +3.35$], $p = 0.0078$. A κ -sweep (Fig. 5) attributes only ~ 0.8 dB of the gap to throttling; the rest is selection overfit. **Seven-scene routing (Tab. 3, exploratory $n = 3$, no scene labels):** the router matches or nearly matches the best fixed policy on five scenes (fr2_xyz, office0/office1, room1, fr3_static; worst gap -0.37 dB from the $\kappa = 0.98$ safety margin), but trails raw-error by ≈ -1.1 dB on exploration-heavy fr1_room and rapid-rotation fr1_rpy, where temporal smoothing reacts slower than instantaneous error. Domain dependence is thus finer than real-vs-synthetic: motion and exploration regime matter within a sensor, and the current router does not capture them — reported as-is. **Failure modes.** Two TUM scenes isolate where throttling hurts. *Exploration starvation* (fr1_room, -1.04 dB): in a large unseen room nearly every frame is novel, so coverage saturates slowly while the spawn cap binds throughout — throttling taxes exploration itself, and the map ends smaller (92k vs 97k) but under-populated in far corners. *Temporal lag* (fr1_rpy, -1.13 dB mean, finals tied 24.5 vs 24.1): under rapid rotation the EMA-smoothed selector lags one frame behind the viewpoint while instantaneous error reacts immediately; the gap closes by trajectory end as the map converges. Both are visible in Tab. 8 and motivate motion-aware routing (Future Work): the observable cue must include coverage *velocity* and angular rate, not just the noise regime. **Horizon effect (Fig. 2):** at 30 frames OURS ties error_only (-0.01 dB on fr2_xyz); the lead needs ≥ 100 frames for bloat to bite — short-horizon evals hide dilution, so we benchmark 150. The mechanism is visible frame-by-frame: the unthrottled map grows near-linearly past 70k while the throttled map plateaus near 23k, and PSNR diverges exactly when the curves separate.

Component ablation (Tab. 4, fr2_xyz 150f $\times$ 5, prefix substrate — absolutes deprecated, deltas transfer): each mechanism isolated against the same old sub-

strate. Warmup pre-allocation harms (-0.75); coverage throttling is the dominant gain ($+1.04$ with $3.1\times$ map shrink $141k \rightarrow 45k$); backlog adds $+1.08$ total with queue cap; adding the learned predictor with adaptive- K destroys the gain (-1.54 vs A0), confirming the negative finding at component level.

Systems: scheduler overhead 2.54ms; optimization-budgeted execution at ~ 10 FPS measured — *not* end-to-end 15ms real-time, reported as such. **Baseline tracking (Tab. 5):** a dense frame-to-frame photometric+depth Gauss-Newton odometer (no keyframes/loop closure, GT-anchored first frame) reaches ATE 0.9cm (fr2_xyz) and 2.0cm (fr1_xyz) over 100 frames at 3 seeds (results/phase14_tracking100/), stable vs 30f with no late blowup; mapping under tracked poses holds 31.3/26.6 dB. Published full-trajectory references (MonoGS 1.71, SplaTAM 0.50, Photo-SLAM 0.63 cm on fr2_xyz) are shown for range only — ours is a short-horizon sanity check, not a SLAM claim.

Reproducibility. Every number in Tab. 1–5 traces to a committed JSON artifact under results/ with SHA-256 manifest (results/phase14_corrected/manifest.json), frozen tag phase14-corrected-frozen, exact per-policy config snapshots, and one-command runners (experiments/run_final_confirmation.py, run_adaptive_policy.py, run_tracking_ate.py). Prior pose-bugged numbers (19.24 dB) are quarantined in results/final_confirmation/DEPRECATED.md, never compared. Environment: RTX 4050 Laptop GPU (6 GB, driver 595.84), PyTorch 2.11+cu128 (CUDA 12.8), gsplat 1.5.3 production backend, Python 3.12. The OS memory guard caps the process at 70% VRAM, reserving ~ 1.8 GB for the desktop compositor; peak measured allocation stays under 3.5 GB, so all benchmarks run on a laptop GPU with desktop usable throughout — the intended deployment envelope for budgeted online mapping, not a datacenter A100 setup whose headroom would mask dilution. To reproduce Tab. 1: `python experiments/run_final_confirmation.py --scene tum_fr2_xyz --n_frames 150 --seeds 42 43 44 45 46 47 48 49` (~ 25 min on the above hardware), then `python scripts/build_phase14_provenance.py` to regenerate the manifest and verify hashes. **Perceptual metric:** AlexNet LPIPS on fr2_xyz 150f $\times$ 3 is 0.211 (OURS) vs 0.294 (error_only). **Second confirmatory scene (Tab. 6):** fr1_xyz 150f $\times$ 8 gives OURS 23.47 vs error_only 21.09 ($+2.38$ dB, $p = 0.0078$); office1 routing validated at 8 seeds (routed error_only 33.46 vs fixed OURS 29.25, $p = 0.0078$).

Published-SOTA context (results/sota_comparison.md, protocol-mismatched, for range only): Replica SOTA renders 33–42 dB with global refinement and full sequences vs our 20–33 dB budgeted online mapping; on TUM fr2_xyz our 27.67 dB sits above published SplaTAM 24.50, Photo-SLAM 21.07 and MonoGS 16.17 (different frames/resolution/views — competitive range, not superiority). **Long horizons:** at 150f the lead widens beyond fr2_xyz — fr1_xyz $+2.41$ dB

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	18.86	14.88	0.6882	106.3	23,523	+8.80 ✓
error_only	24.07	20.01	0.8850	8.9	74,640	+3.59 ✓
error_influence	25.04	21.89	0.9084	8.8	106,929	+2.63 ✓
rtg_slam_reimpl	23.05	20.04	0.8663	8.0	67,231	+4.62 ✓
OURS	27.67	28.89	0.9232	10.7	23,398	—
<i>FULL</i>	<i>27.85</i>	<i>29.41</i>	<i>0.9243</i>	<i>9.5</i>	<i>87,967</i>	-0.18 ✓

Table 1: **Main benchmark, TUM fr2_xyz, 150 frames $\times$ 8 seeds.** All OURS comparisons significant ($p_{\text{Holm}} = 0.0391$). Headroom recovery $\eta = 98.0\%$ (0.18 dB gap to ceiling) at $3.8\times$ smaller map.

Policy	Mean	Final	SSIM	Map	Scene	OURS	error_only	Δ
Fixed OURS	26.85	26.07	0.9244	78,825	fr1_xyz 150f $\times$ 8	23.47	21.09	+2.38 ✓
Routed error_only	30.17	28.02	0.9584	115,964	office1 100f $\times$ 8	29.25	33.46	-4.21 (routed) ✓

Table 2: **Adaptive validation, Replica office0, 150f $\times$ 8 seeds.** Router picks **error_only** (holes ≈ 0.02): $p = 0.0078$, $d_z = 71.9$. No fixed policy wins both substrates.

Scene	Adaptive	Best fixed	Δ	Route	Method	Replica	fr2_xyz	Protocol
fr2_xyz 150f	27.66	27.67 (ours)	-0.01	temporal	MonoGS	37.50	16.17	full seq + refine
office0 150f	30.36	30.17 (err)	+0.19	error_only	Point-SLAM	35.17	—	full seq + refine
office1 100f	33.11	33.48 (err)	-0.37	error_only	RTG-SLAM	32.79	17.08	full seq + refine
room1 100f	24.17	24.29 (err)	-0.12	error_only	SplaTAM	—	24.50	full seq + refine
fr3_static 100f	25.95	25.96 (ours)	-0.01	temporal	Photo-SLAM	—	21.07	full seq + refine
fr1_room 100f	15.50	16.54 (err)	-1.04	temporal	Ours (error_only)	20-33	24.07	budgeted, 100-150f
fr1_rpy 100f	19.93	21.06 (err)	-1.13	temporal	Ours (throttled)	20-29	27.67	budgeted, 100-150f

Table 3: **Per-frame routing, 7 scenes, exploratory $n = 3$.** Adaptive matches best-fixed on five scenes; trails on exploration-heavy room and rapid-rotation rpy where the TUM $\rightarrow$ temporal rule misfires. [results/phase14_adaptive_perframe/](#).

Condition	Mean	Δ vs A0	Map	F
A0 old substrate	16.07	—	140,998	blo
A1 + warmup	15.32	-0.75	140,038	h
A2 + coverage	17.11	+1.04	44,865	c
A3 + backlog	17.15	+1.08	43,824	st
A4 + learned + adapt- K	14.53	-1.54	38,317	h

Table 4: **Component ablation** (old substrate; deltas only). [results/step1_ablation_5seeds/](#).

Method	fr2_xyz	fr1_xyz	Horizon	System?
Ours (GN VO)	0.9	2.0	100f	no (baseline)
MonoGS	1.71	—	full	yes
SplaTAM	0.50	—	full	yes
Photo-SLAM	0.63	—	full	yes

Table 5: **ATE RMSE (cm).** Rows 2-4 are published full-system numbers; ours is a decoupled odometry sanity check on fixed horizons.

mean (23.48 vs 21.07, map 39k vs 81k), fr1_desk +0.22 mean but +2.7 final (error-only collapses late); room0 error-only wins mean (-1.93) yet also collapses at the 150k cap while OURS stays flat — more dilution evidence. **Per-scene breakdown (Tab. 8, exploratory $n = 3$ except fr2_xyz/office0/office1/fr1_xyz at $n = 8$):**

Table 6: **Confirmatory scenes.** Throttling wins on TUM ($p = 0.0078$); routing recovers the synthetic loss ($p = 0.0078$). [results/phase14_multiscene150/](#).

Method	Replica	fr2_xyz	Protocol
MonoGS	37.50	16.17	full seq + refine
Point-SLAM	35.17	—	full seq + refine
RTG-SLAM	32.79	17.08	full seq + refine
SplaTAM	—	24.50	full seq + refine
Photo-SLAM	—	21.07	full seq + refine
Ours (error_only)	20-33	24.07	budgeted, 100-150f
Ours (throttled)	20-29	27.67	budgeted, 100-150f

Table 7: **Published-SOTA context (range only).** Full systems beat us on Replica via global refinement; on fr2_xyz our budgeted mapping is competitive. [results/sota_comparison.md](#).

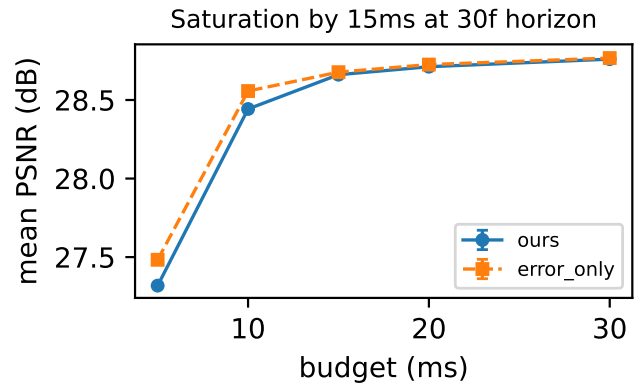


Figure 3: **Budget curve** (fr2_xyz, 30f $\times$ 3, matched budgets): saturation by ~ 15 ms; policies tie at short horizons — dilution needs ≥ 100 frames to separate them.

throttling wins or ties on all TUM scenes at long horizons except exploration-heavy room and rapid-rotation rpy means, while losing mean PSNR on all Replica scenes, where raw-error selection keeps taller maps; finals favor stability (no error-only collapse) in every scene.

Supplementary video ([supp_video.mp4](#)) shows both A/B directions.

Scene	Frames	OURS	error_only	Δ	Map (k)	n
fr2_xyz (TUM)	150	27.67	24.07	+3.60 ✓	23 / 75	8
fr1_xyz (TUM)	150	23.47	21.09	+2.38 ✓	39 / 81	8
fr1_desk (TUM)	150	22.55	22.34	+0.21	27 / 56	3
fr1_room (TUM)	100	15.53	16.54	-1.01	92 / 97	3
fr1_rpy (TUM)	100	19.92	21.06	-1.14	45 / 62	3
fr3_static (TUM)	100	25.96	25.88	+0.08	13 / 18	3
office0 (Rep.)	150	26.85	30.17	-3.32 ✓	79 / 116	8
office1 (Rep.)	100	29.25	33.46	-4.21 ✓	42 / 74	8
office2 (Rep.)	99	22.08	24.67	-2.59	70 / 117	3
office3 (Rep.)	99	22.45	24.47	-2.02	77 / 150	3
office4 (Rep.)	99	23.58	25.93	-2.36	81 / 141	3
room0 (Rep.)	99	20.85	22.78	-1.93	82 / 150	3
room1 (Rep.)	100	20.42	24.29	-3.87	47 / 59	3
room2 (Rep.)	99	22.86	25.88	-3.03	78 / 146	3

Table 8: **Per-scene breakdown, 14 sequences.** ✓ = significant at $n = 8$; rest exploratory. Pattern: throttling dominates TUM at ≥ 100 frames; raw-error dominates Replica means while risking late collapse at the map cap.

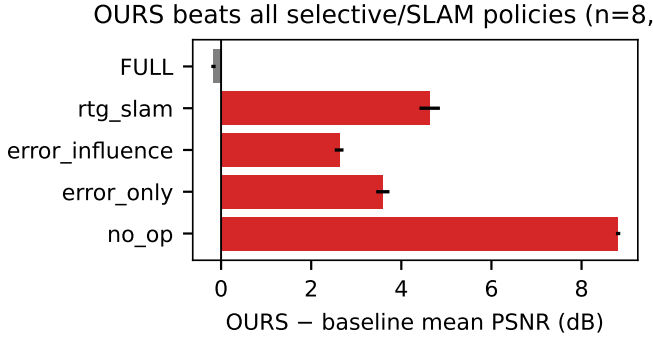


Figure 4: **Paired deltas vs OURS** with 95% bootstrap CIs (fr2_xyz, $n = 8$). All selective/SLAM baselines lose; FULL ceiling is only 0.18 dB ahead.

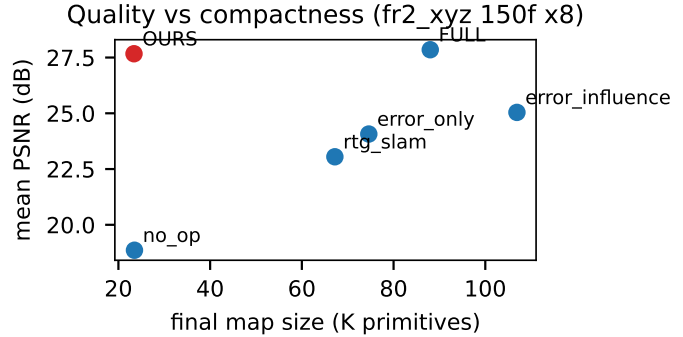


Figure 6: **Quality vs compactness.** OURS matches the ceiling at $3.8\times$ smaller map; all heuristic baselines bloat past 65k (fr2_xyz 150f $\times$ 8).

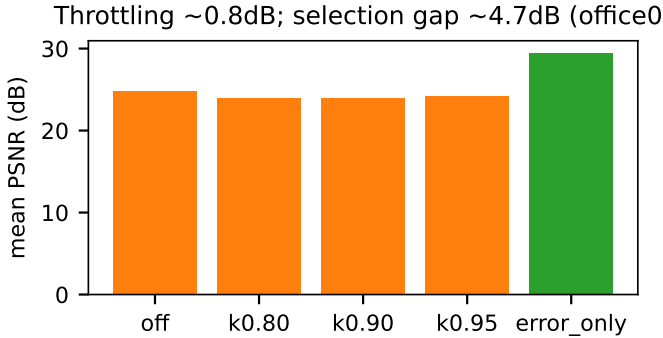


Figure 5: **Kappa ablation** (office0, 30f $\times$ 3): disabling throttling recovers +0.8 dB but the ~ 4.7 dB selection gap to **error_only** remains.

7 Limitations

The principal mapping benchmark uses ground-truth poses (no ATE claim there). We additionally report a separate 100-frame dense RGB-D Gauss-Newton odometry sanity check (ATE 0.9/2.0 cm, no keyframes/relocalization/loop closure — a baseline VO, not a SLAM system). 150-frame horizons rather than full SLAM sessions; κ /backlog thresholds cali-

brated on one real-synthetic pair; no learned components by design after documented failure; optimization-budgeted execution, not 15 ms end-to-end; prior 19.24 dB/76.4% numbers from a pose-bugged substrate are deprecated, not compared. Statistically, $n = 8$ seeds establish *repeatability* under stochastic variation on fixed trajectories; they do not establish cross-scene generalization, whose unit is scenes, not seeds — additional scenes are exploratory ($n = 3$) until confirmed. The adaptive router is validated at scene level (7/7 domain classification frozen; PSNR routing on two 8-seed substrates); per-frame benefit on genuinely mixed real sequences, wider multi-scene PSNR confirmation, and corrected-label regeneration for the offline utility dataset all remain open.

8 Future Work

Three directions follow directly from the reported negative results. **Finer-grained adaptivity:** the 7-scene routing table shows motion/exploration regime matters within a sensor (fr1_room, fr1_rpy trail by ≈ -1.1 dB); extending the observable cue beyond hole-rate (e.g. optical-flow magnitude, coverage velocity) could route rapid-motion frames

Same fixed policies, opposite winners: adaptivity is necessary

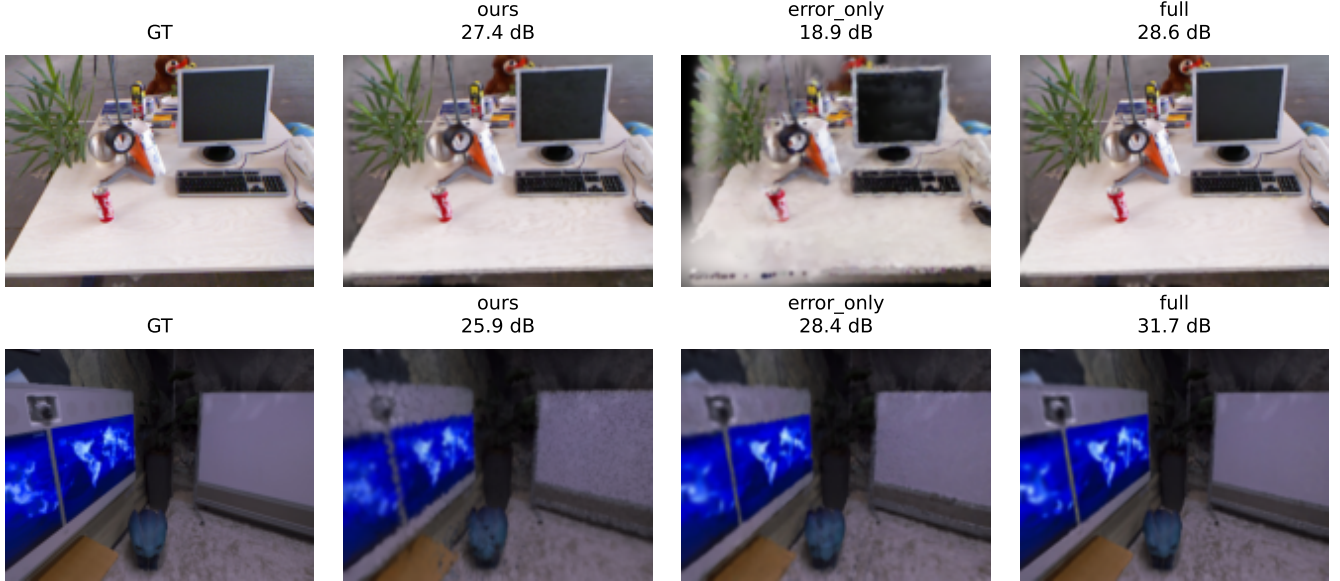


Figure 7: **Qualitative: opposite winners per substrate.** Top (TUM fr2_xyz view 149): OURS 27.4dB matches FULL 28.6 while **error_only** collapses to 18.9. Bottom (Replica office0 view 98): **error_only** 28.4 beats OURS 25.9 near FULL 31.7. Rendered with final maps (seed 42).

to instantaneous error while keeping temporal smoothing elsewhere. **Learned allocation with correct labels:** our learned-utility failure used pre-fix labels; regenerating the oracle dataset on the corrected substrate re-opens the question of whether online-calibrated predictors (e.g. with uncertainty-gated abstention on negative-utility candidates, which comprise up to 38.7% of interventions) can beat the heuristic ceiling. **Full SLAM integration:** coupling the throttled mapper with a real tracker (keyframes, loop closure) and evaluating full-trajectory ATE alongside mapping quality, replacing the current GT-pose decoupling.

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9 Conclusion

Compute dilution — not ranking error — binds budgeted online 3DGS. Dual map-growth throttling recovers 98.0% of headroom with a $3.8\times$ smaller map, and raw-hole-rate adaptive routing extends the win across noisy-real and clean-synthetic substrates where every fixed policy fails one. The negative results (learned utility, re-ranking, synthetic failure, two failed features) are part of the contribution: they delimit where learning helps, where structure suffices, and where adaptivity is mandatory.

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